Data and Signals: Theoretical Channel Capacity

Coverage: Analog vs Digital Signals, Bandwidth, Transmission Impairments (Attenuation, Noise), Nyquist Rate vs Shannon Capacity

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1. Signals & Impairments

• Bandwidth B = f_max - f_min. Attenuation dB = 10 * log10(P2/P1). SNR_dB = 10 * log10(SNR).

2. Channel Capacity

• Nyquist (Noiseless): C = 2 * B * log2(L) bps. Shannon (Noisy): C = B * log2(1 + SNR) bps.

■ Worked Problem: Shannon & Nyquist Capacity

• Given B = 3000 Hz, SNR_dB = 31.62 dB -> SNR = 1452.1.

• Shannon C = 3000 * log2(1453.1) = 31.5 kbps. Nyquist levels: 31,515 = 6000 * log2(L) -> L = 38 levels.

■ KTU University High-Yield Exam Questions

• Q: State and derive Nyquist and Shannon capacity formulas with a numerical problem. (8 Marks)